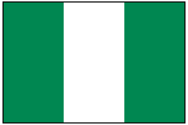
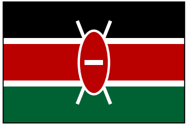
Ethiopia



Nigeria



Kenya

Confidently Wrong

Calibration Risks and Safe Abstention in LLMs for African Languages

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A team from Ethiopia, Nigeria, and Kenya · Global South AI Safety Hackathon, June 2026

The problem

- LLMs are increasingly used in African languages, but their reliability there is largely **unmeasured**.
- The failure we study is being **confidently wrong**: a wrong answer delivered with high confidence.
- A miscalibrated model, whose confidence does not track its accuracy, cannot know when to **abstain**.

Why it matters for African AI safety

- These models answer **high-stakes** questions: health, legal, educational, civic.
- Users often **cannot verify** an answer in their own language.
- If calibration holds in English but breaks in African languages, the populations **least served** are the most exposed to confident misinformation.

Methods

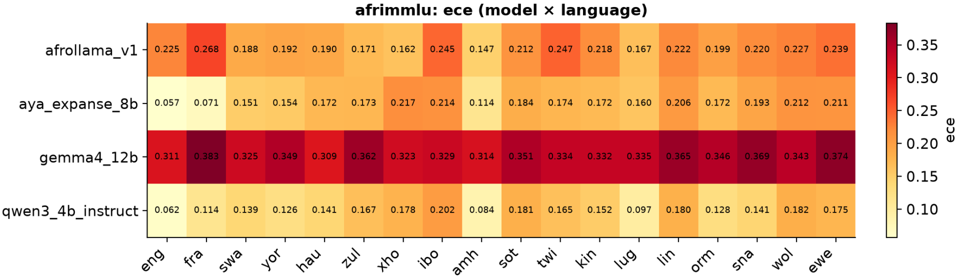
- **Four open models:** Qwen3-4B-Instruct, Gemma-4-12B, Aya-Expanse-8B, AfroLlama-V1.
- **Two benchmarks:** Uhura-TruthfulQA (truthfulness) and AfriMMLU (knowledge).
- **Scoring:** length-normalised answer-text log-probability (robust to instruction-tuned models).
- **Metrics & interventions:** ECE, overconfidence, AUROC, AUARC; temperature scaling, abstention, few-shot.

Result: calibration degrades in African languages

- ECE and overconfidence are markedly higher in African languages, across models and tasks.
- Models are most overconfident exactly where they are least accurate.

Model (Uhura)	ECE en	ECE afr
Qwen3-4B	0.07	0.17
Gemma-4-12B	0.21	0.26
Aya-Expanse-8B	0.08	0.20

Result: calibration error by language

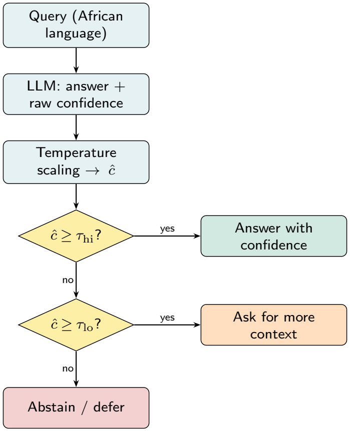


ECE is low in English and French and rises across African languages.

Result: mitigations are unequal

- **Temperature scaling** is the clear winner: per-language recalibration cuts ECE to ≈ 0.03, with no retraining.
- **Abstention** helps less: the uncertainty signal is weak (AUROC ≈ 0.55).
- **Few-shot** prompting helps three models but **backfires on Gemma-4-12B**.
- **Bigger is not safer:** the 12B model is the worst calibrated.

Deployment framework



Recalibrate first, then route on calibrated confidence; thresholds τ set per language.

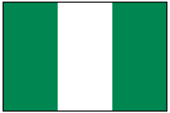
Limitations & future work

- Accuracies sit near chance on these hard tasks; ECE gaps may partly reflect translation quality.
- Abstention is bounded by a weak uncertainty signal, so recalibration is the more reliable lever.
- **Next:** reasoning and closed models; in-context calibration at scale; native-speaker audits and human-grounded abstention.

Takeaways

- Open LLMs are **confidently wrong** in African languages: less accurate, more overconfident.
- Cheap, post-hoc **temperature scaling** is a deployable safety lever today.
- Pair recalibration with **thresholded abstention** for safer real-world use.

Ethiopia



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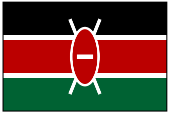

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Thank you

github.com/rexsimiloluwah/aims-team-apart-ai-safety-hackathon